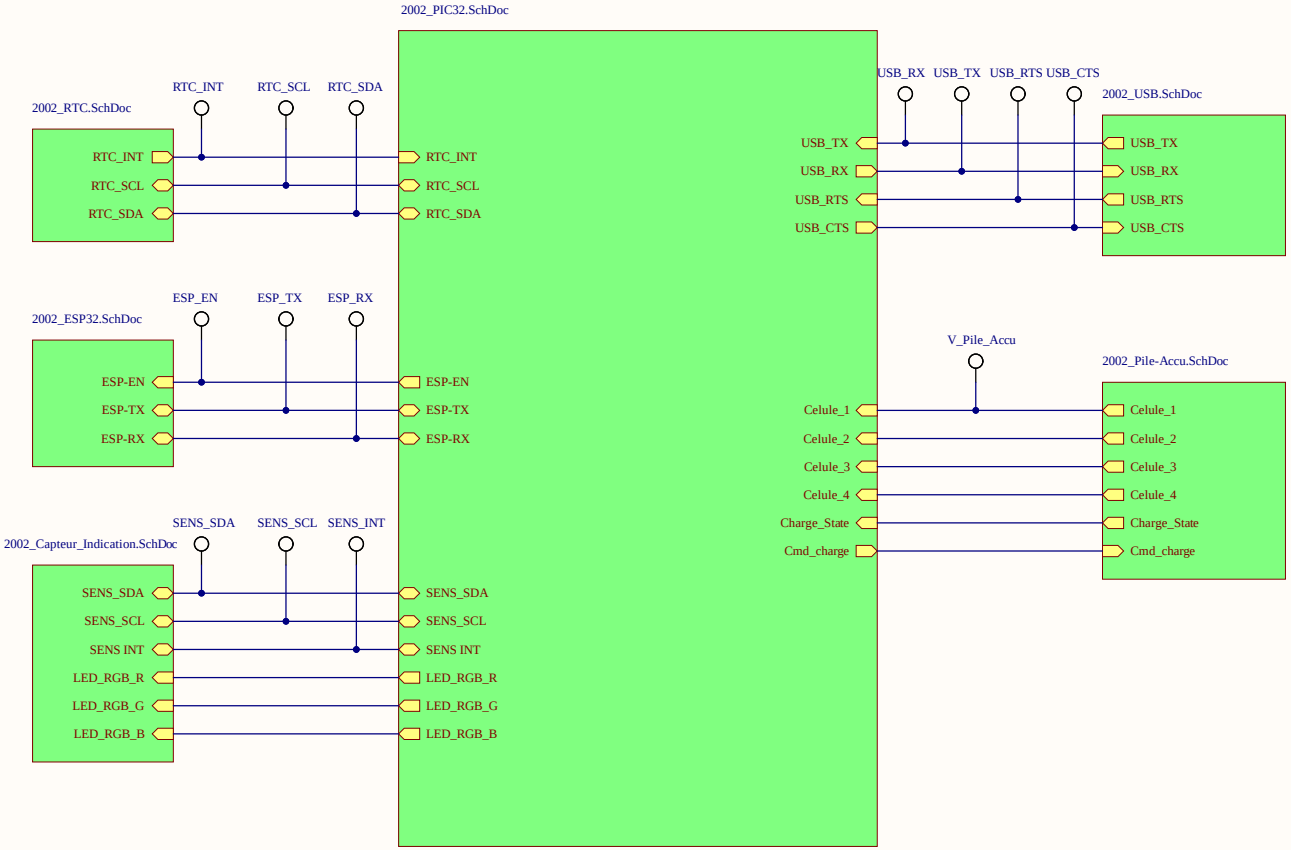
2002 Capteur Wi-fi



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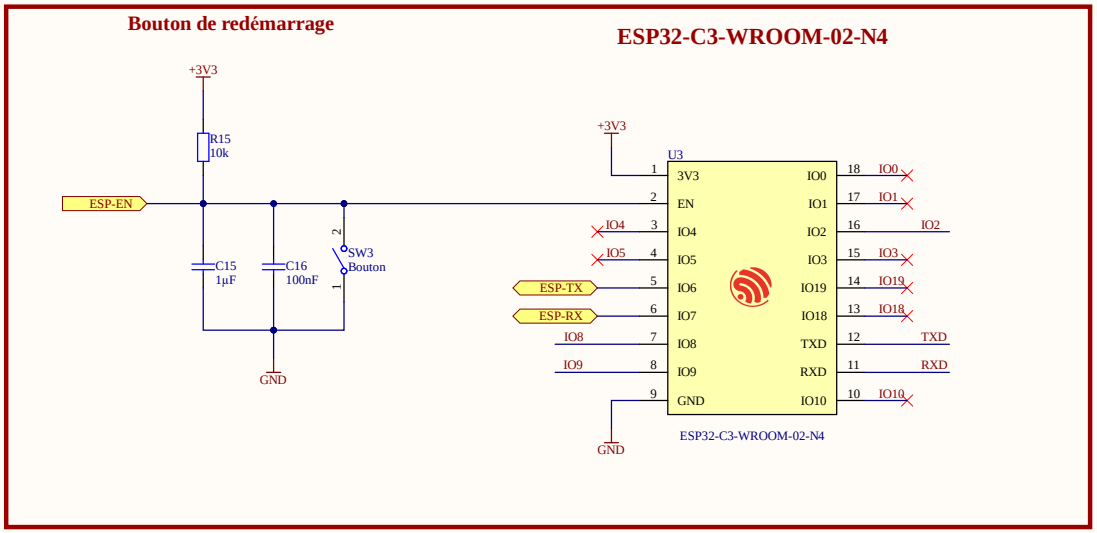
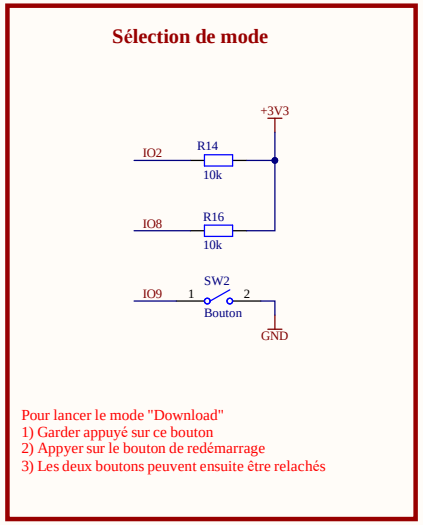
No :

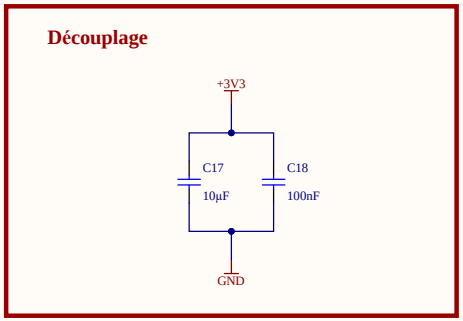
Nb feuilles 7 | Feuille n° 1

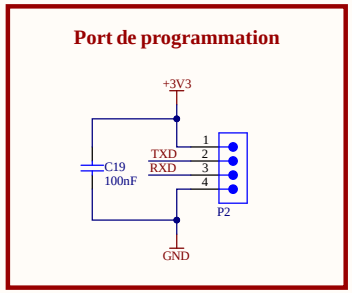
Chemin : H:\2002_C\hard\2002_C\Schéma\2002_SchemaBloc.SchDoc

ESP32









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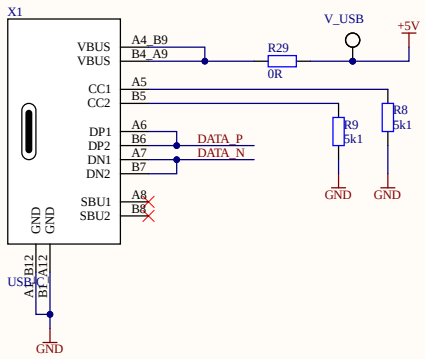
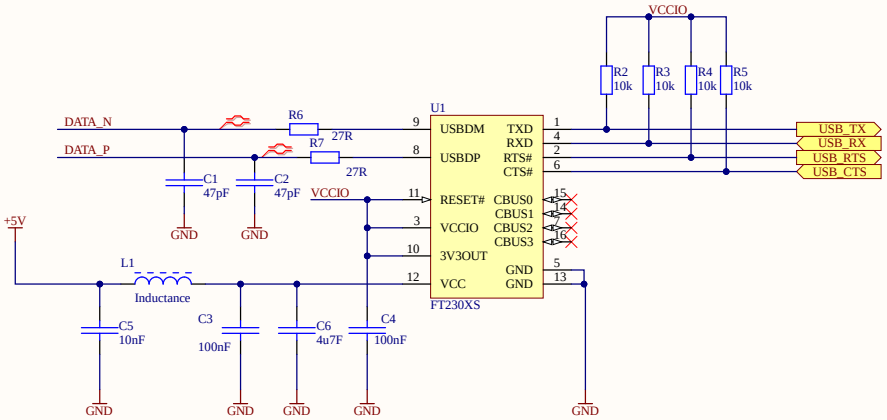
230VAC / Alimentation

USB



Convertisseur USB/UART



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No :

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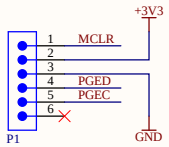
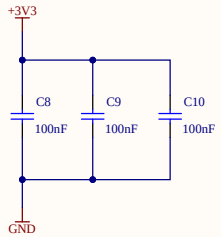
PIC32

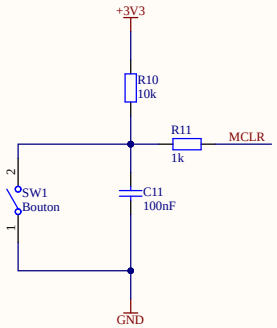
Port de programmation



Découplage



Bouton de redémarrage



PIC32MX130F256D

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No :

Nb feuilles 7 | Feuille n° 4

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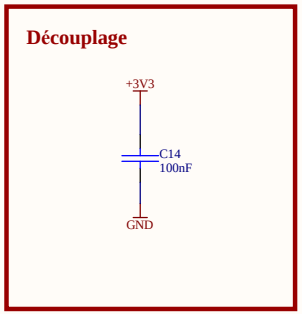
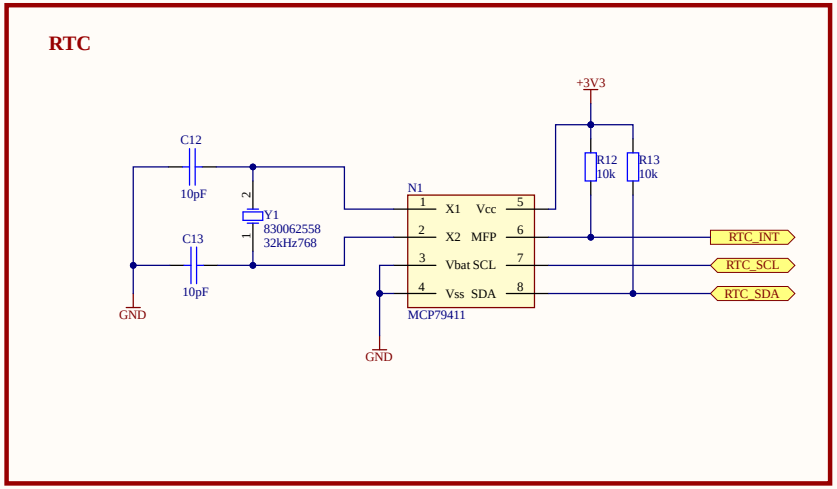
RTC





Fichier : 2002_RTC.SchDoc		Modif.	a
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No :

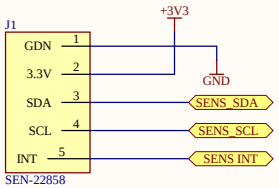
Nb feuilles 7 | Feuille n° 5

Chemin : H:\2002_C\hard\2002_C\Schéma\2002_RTC.SchDoc

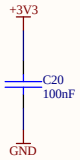
Capteur

Indication

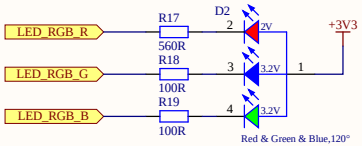
Module Sparkfun Capteur (BM280/ENS160)



Découplage



LED RGB



Fichier : 2002_Capteur_Indication.SchDoc		Modif.	a
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			c
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No :

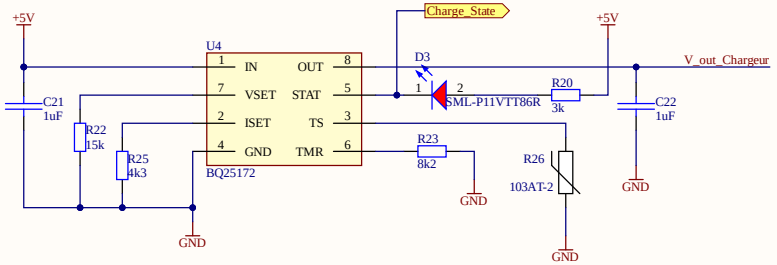
Nb feuilles 7

Feuille n° 6

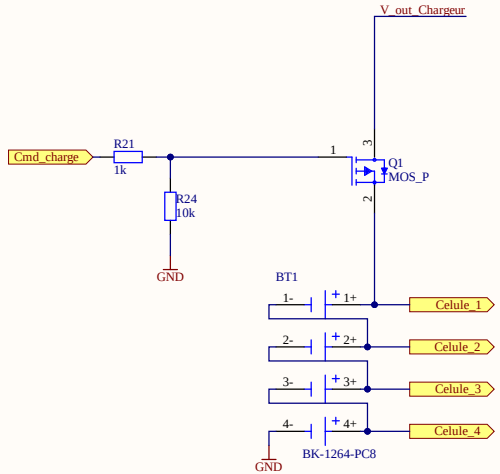
Chemin : H:\2002_C\hard\2002_C\Schéma\2002_Capteur_Indication.SchDoc

Pile/Accu

Controleur de charge

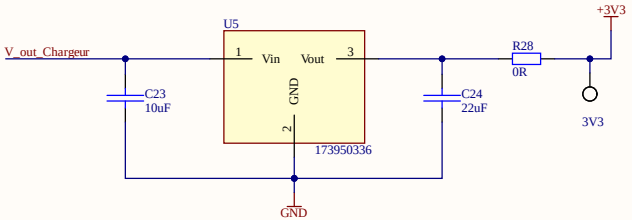


Commande de charge



Si nous détectons un problème, nous allons stopper la charge à l'aide d'un MOSFET à canal P qui contrôle la charge des piles. Lorsque l'action est désactivée, la connexion sera OFF. Si le système s'arrête brusquement, le MOSFET basculera en position ON.

Regulateur 3.3V



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ETML-ES Avenue Recordon 1 1004 Lausanne Switzerland	Projet : 2002_CapteurMeteoWifi.PrjPcb	No :	
		Nb feuilles 7	Feuille n° 7
Chemin : H:\2002_C\hard\2002_C\2002_Pile-Accu.SchDoc			

Designator	Comment	Description	Quantity
R28, R29	0R	Resistor smd, Resistor	2
R11, R21	1k	SMD Resistor, Resistor smd	2
C15, C21, C22	1uF	Capacitor smd	3
R20	3k2	Resistor smd	1
P2	4 pin header	WR-PHD Pin Header, THT, Vertical, pitch 2.54mm, 1 Row, 4P	1
R25	4k3	Resistor smd	1
C6	4u7F	0805 Condensator	1
R8, R9	5k1		2
P1	6 Pin header	WR-PHD Pin Header, THT, Vertical, pitch 2.54mm, 1 Row, 6P	1
R23	8k2	Resistor smd	1
R2, R3, R4, R5, R10, R12, R13, R14, R15, R16, R24	10k	SMD Resistor, Resistor smd	11
C5	10nF	0805 Condensator	1
C12, C13	10pF	0805 Condensator	2
C7, C17, C23	10uF	0805 Condensator, Capacitor smd	3
R22	15k	Resistor smd	1
C24	22uF	Capacitor smd	1
R6, R7	27R	SMD Resistor 0805, 330Ω	2
C1, C2	47pF	Condensator	2
C3, C4, C8, C9, C10, C11, C14, C16, C19, C20	100nF	0805 Condensator, Capacitor smd	10
C18	100nF16	Capacitor smd	1
R18, R19	100R	SMD Resistor	2
R26	103AT-2	thermistor	1
R17	560R	SMD Resistor	1
3V3, ESP_EN, ESP_RX, ESP_TX, GND1, GND2, RTC_INT, RTC_SCL, RTC_SDA, SENS_INT, SENS_SCL, SENS_SDA, USB_CTS, USB_RTS, USB_RX, USB_TX, V_Pile_Accu, V_USB	5019	Test point smd	18
U5	173950336	DC DC CONVERTER 3.3V	1
Y1	830062558	Crystals WE-XTAL 32.7680kHz 20ppm 2.0 x 1.2mm	1
BT1	BK-1264-PC8	Battery Holder (Open) AAA 4 Cell PC Pin	1
SW1, SW2, SW3	Bouton	Tactile Switches WS-TASV Tact Switch	3
U4	BQ25172	Linear Battery Charger for 1- to 6-Cell NiMH Batteries	1
U3	ESP32-C3-WROOM-02-N4	WiFi Modules (802.11) (Engineering Samples) SMD module, ESP32-C3, 4MB SPI flash, PCB antenna, -40 C +85 C	1
U1	FT230XS	USB to Basic UART IC, 2.97 to 5.5 V, -40 to 85 degC, 16-Pin SSOP, RoHS	1
L1	Inductance	MI0805K601R-10	1
N1	MCP79411	RTC	1
Q1	MOS_P		1
U2	PIC32MX130F256D	Integrated Circuit	1
D2	Red & Green & Blue,120°	DEL standard - CMS RGB 0603 SMD	1
J1	SEN-22858	BME280/ENS160	1
D3	SML-P11VTT86R	WL-SMCW Mono-color Chip LED Waterclear, size 0805, Super Red, 1.9V, 140°	1
X1	USB-C	U262-161N-4BVC11	1